

Dependence of the foaming window of a polystyrene/poly(methyl methacrylate) blend on structural evolution driven by phase separation

Wei Liu, Yongyan Pang*, Minghui Wu, Kangpei Qin, Wenge Zheng**

Ningbo Key Laboratory of Polymer Materials, Key Laboratory of Bio-based Polymeric Materials Technology and Application of Zhejiang Province, Polymers and Composites Division, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo, Zhejiang Province, 315201, China

HIGHLIGHTS

- Foaming behavior is affected by structural evolution via phase separation.
- Phase domain size plays a significant role in affecting cell structure.
- Upper foaming temperature is vulnerable to large domain and high pressure.
- Foaming window is narrowed for combination of large domain and high pressure.
- Viscoelasticity difference of two phases affects foaming behavior significantly.

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ABSTRACT

The objective of the present work was to study the dependence of the foamability and the foaming window of a polystyrene/poly (methyl methacrylate) (PS60/PMMA40) blend on the structural evolution driven by phase separation. A compression molding machine was used to prepare the blend samples thermally annealing for various times. The phase contrast optical microscope (PCOM) and scanning electron microscopy (SEM) were used for phase morphological change study. The samples were foamed via a batch foaming process, and SEM was used for cell structure observation. The desorption curves of the blend samples were studied to show the dependence of the gas desorption diffusion coefficient on the annealing. It was found that the phase domain size was increased in a wide range with the annealing, and that the foaming behavior was significantly affected by the structural evolution for the blend with big phase domains and the saturation under high pressures. The gas desorption diffusion coefficient was increased for the interface with the phase domain, which resulted in the decreased expansion ratio after foaming, with varying reduction degrees for varying saturation pressures. Further, the suitable foaming temperature range was studied for the blend samples annealing for a short and a long period and saturated under each pressure. It was found that both the upper and the lower foaming temperatures were decreased with the saturation pressure, and the upper one was more vulnerable to the higher pressures for the blend samples after long-period annealing, leading to the much narrower suitable foaming temperature window. It was believed that the difference in viscoelasticity of the dispersed phase and the matrix was responsible for this phenomenon.

1. Introduction

Since the development of microcellular foaming techniques in 1980s, polymer foaming with green foaming agents has been of great interests in both academic and industrial fields, especially in response to the appeal of energy saving and environmental friendliness for a sustainable development path [1,2]. Polymer blending is a simple, practical and economical means for obtaining materials with good properties that are

absent in single-polymer systems. To meet the end-use demands in applications, polymer blends are usually required to fabricate foam materials with desired cell structures and properties [3,4].

Compared to the single-polymer systems, there are more influential factors in affecting the foaming behavior and cell structures of binary polymer blends, such as the compositions, the phase morphologies, the processing conditions and the different foaming characteristics of the two components, etc [5–12]. Among these factors, phase morphology

* Corresponding author. 1219 Zhongguan West Road, Ningbo, Zhejiang Province, 315201, China.

** Corresponding author. 1219 Zhongguan West Road, Ningbo, Zhejiang Province, 315201, China.

E-mail addresses: yongyan.pang@nimte.ac.cn (Y. Pang), wgzhen@nimte.ac.cn (W. Zheng).

plays an important role in affecting the cell structures. The phase morphology is usually impacted by the compositions, the viscosity ratios of the two components, the interfacial compatibilization and the processing conditions, etc [5,7,9]. Huang et al. [5] reported that the cell structures of polypropylene/polystyrene (PP/PS) blends are tunable via varying the compositions. Zhai et al. [6] studied the foaming behavior of PP/PS blends with the interfacial compatibilization of PS-grafted PP copolymers (PP-g-PS), and found that the interfacial compatibilization favors the increased cell density and expansion ratio. Zhang et al. [7] found that the effect of the compatibilization of linear low density polyethylene/polystyrene (LLDPE/PS) blends with polyethylene-g-polystyrene (PE-g-PS) on the foamability was more prominent for the blends with co-continuous structures. Han et al. [9] studied the dependence of cell structures of polystyrene/poly (methyl methacrylate) (PS/PMMA) blends of different viscosity ratios on the phase morphology via taking samples out from the extruder at various length-to-diameter (L/D) ratios. Wang et al. [10] improved the cell structures of PP/PS blends via application of supercritical CO₂ (scCO₂) as the processing medium to improve the dispersion of the minor phases.

Till now, from the qualitative perspective, it has been well accepted that the phase morphology plays a significant role in determining the foaming behavior and the cell structures in a binary polymer blend. However, from the quantitative perspective, it is still not clear how the phase domain size affects the foamability and thus the foaming window of a polymer blend. To solve this scientific problem, the change of the phase domain size in a wide range in a binary polymer blend of the same composition is required to be tunable and precisely controlled. Utilization of a processing means to tune the phase morphology is believed to be most likely a feasible approach. Based on the previous literature [9,12], the application of screws with varying L/D ratios in extrusion blending can be used to tune the phase morphology, but this means is usually limited as it involves design of screw of different L/D ratios. In addition, application of scCO₂ as the processing medium via continuous extrusion foaming could also alter the phase morphology of polymer blends [10]. However, it requires that the higher-viscosity component in the blend has a higher affinity of scCO₂ to decrease the phase domain size, and if the higher-viscosity component has a lower affinity of scCO₂, the phase domain will otherwise increase. Consequently, this processing means is also limited by good matching viscosity of the components with the solubility of CO₂ in them to precisely control the phase morphology from one side, and usually by lacking of the facilities from the other. Hence, a feasible processing approach is still demanded that is easy to be applied to tune the phase domain size in a wide range.

Thermal annealing at melt state can provide a feasible approach that is easy to conduct to prepare polymer blends of the exact same composition with gradually increased phase domain in a broad range via morphological evolution in time scale driven by phase separation [13–18]. Numerous publications have presented structural evolution and phase separation mechanisms of binary polymer blends as fundamental studies from the thermodynamic perspective [13,14,18]. In addition, Pang et al. [13,14] reported that the evolution of phase structure driven by phase separation after thermal annealing could exert a significant impact on the mechanical properties of polymer blends, depending on the structural change during the annealing. The previous findings are supportive for thermal annealing at melt state as a feasible approach to provide the morphological evolution to quantitatively build the relationship between phase domain size and the foaming window of binary polymer blends.

In this study, a typical immiscible binary polymer blend of PS/PMMA with a weight ratio of 60/40 was employed to study the effect of morphological evolution on the foaming window. First, the structural evolution at melt state was investigated using phase contrast optical microscope (PCOM) and scanning electron microscopy (SEM). Second, SEM was used to study the cell structures of the PS/PMMA blend foams prepared after annealing for different times and saturation under

various pressures. Third, the desorption diffusion coefficient and the expansion ratio of the blends were studied as a function of the annealing time. Fourth, the foaming temperature window under different saturation pressures was identified to reveal the effect of the phase domain size and the saturation pressure on changing the foaming window.

2. Experimental section

2.1. Materials

General purpose polystyrene (PS, GPPS251) resin with MFI of 2.4 g/10 min at 200 °C/5 kg, a weight average molecular weight of 1.4×10^5 g/mol and a polydispersity of 2.05 was provided by Shanghai Secco Petrochemical Co., Ltd (China). Poly (methyl methacrylate) (PMMA, CM207) with MFI of 8.0 g/10 min at 230 °C/3.8 kg, a weight average molecular weight of 4.3×10^4 g/mol and a polydispersity of 1.78 was purchased from Zhenjiang Chi Mei Chemical Co., Ltd (China). The physical foaming agent, i.e., commercial grade carbon dioxide (purity 99%) was provided by Ningbo Wanli Gas Corporation.

2.2. Sample preparation

The PS and PMMA pellets were dried in a vacuum oven at 70 °C for 12 h before use, and then were premixed at a weight ratio of 60/40 in a plastic bag. A co-rotating twin screw extruder (Brabender GmbH & Co KG, Plastic-Corder Lab-Station) was employed to blend the premix subsequently at the speed of 25 rpm. The temperatures were set at 210, 220, 220, 220, 220 and 220 °C from the feed to the die zones. The blending was performed twice to achieve better dispersion. Thermal annealing of the PS/PMMA blend was conducted using a compression molding machine (Huzhou Shuangli Automation Technology, XLB50-D) at 220 °C and 10 MPa for 2, 7, 15, 30, 60, 100 min to prepare samples with the dimensions of 75 mm × 75 mm × 1 mm for foaming and other tests.

2.3. Observation of phase structure evolution

The Olympus BX53 optical microscope connected with a Linkam LTS350 hot stage was used to study the evolution of phase morphology of the PS60/PMMA40 blend. A thin film with a thickness of ca. 50 μm was used for the phase contrast optical microscopy (PCOM), prepared via compression molding at 220 °C for 1 min under a pressure of 10 MPa. The film was placed on a hot stage at room temperature, and then was rapidly heated to 220 °C. During the phase morphology evolution, the phase contrast micrographs were taken with the camera connected to the optical microscope at the annealing time of 2, 7, 15, 30, 60, 100 min, respectively.

The morphological change was also studied after the PS60/PMMA40 blend samples were thermally annealed for various times using a compression molding machine. A Zeiss EVO18 scanning electron microscope (SEM) was employed to study the structural evolution after a freeze fracture of the samples in liquid nitrogen and deposition of a layer of gold on the surface.

2.4. Batch foaming

The batch foaming approach was used in this study as it allows for precise structural control before saturation and foaming [19]. After high-temperature annealing, the samples were kept in CO₂ atmosphere in a high pressure vessel at room temperature for 48 h, and the saturation pressure was set at 1.5, 2, 3, 3.5 and 4 MPa, respectively. After saturation, CO₂ was released immediately, and all the samples were transferred into the glycerol bath pre-set at a series of temperatures to foam for 15 s. Finally, all the samples were transferred from glycerol bath into ice water to stabilize the cell structure.

2.5. Cell structure investigation

The cell structures of the PS60/PMMA40 foam samples were studied using the Zeiss EVO18 SEM at an operating voltage of 20 kV after the samples were cryogenically fractured in liquid nitrogen followed by a layer of gold was sputtered on the fractured surface.

The expansion ratio (φ) of the PS60/PMMA40 foam samples was calculated based on equation (1) [10]:

$$\varphi = \frac{\rho}{\rho_f} \quad (1)$$

The mass density of the unfoamed (ρ) and foamed (ρ_f) samples are obtained through the water displacement method according to ISO 1183–1987.

2.6. Desorption curve and desorption diffusion coefficient

The desorption curves of PS60/PMMA40 blend samples were studied to reveal the effect of the annealing on the desorption diffusion coefficient. The sheet samples annealed for different times were enclosed in a high pressure vessel and were saturated in CO₂ atmosphere at 3 MPa at room temperature for 48 h. Then following a rapid venting of the vessel, the samples were transferred from the vessel to a digital balance (accuracy at 0.1 mg) to record the mass loss versus time. The desorption curve was obtained by plotting $(M_t - M_0)/M_0$ versus $t^{1/2}/L$ [20]. M_0 is the initial sample weight, M_t is sample weight with absorbed CO₂ at the desorption time t , and L is the sample thickness.

Based on the desorption curves, the desorption diffusion coefficient (D_d) of the blend samples annealing for different times can be obtained via equation (2) [21]:

$$\frac{M_\infty - M_t}{M_\infty} = \frac{4}{L} \sqrt{\frac{D_d t}{\pi}} \quad (2)$$

where M_∞ is the saturated amount of CO₂ under the saturation condition and can be obtained by extrapolating the desorption curve to $t = 0$, M_t is the dissolved amount of CO₂ at the desorption time t , and L is the sample thickness. Through linear line fitting of $(M_\infty - M_t)/M_\infty$ as a function of $t^{1/2}/L$, the value of the slope is obtained as $4(D_d/\pi)^{1/2}$. Through calculation, the desorption diffusion coefficient D_d can be obtained.

3. Results and discussion

3.1. Structural evolution of PS60/PMMA40 blends during annealing

It is known that the phase morphology of polymer blends plays an important role in determining the foaming behavior and the cell structures [8,22–24]. The phase morphology evolution of PS60/PMMA40 blends annealed at 220 °C for various times is shown in Fig. 1. The PCOM (upper row of Fig. 1) showed that the PMMA rich phase domains coarsened and grew gradually with the annealing. Within annealing of 7 min, the phase morphology of PS60/PMMA40 was co-continuous structure. After annealing for 15 min, some of the phase domains evolved into sea-island structure. With prolonged annealing, the dispersed phase domains turned circular and bigger in size, driven by interfacial tension. Based on the previous studies [25,26], PS/PMMA

blend is a typical immiscible binary polymer blend, which has an upper critical solution temperature (UCST). In this study, the structural evolution during coarsening follows a spinodal decomposition phase separation mechanism, and is in good agreement with other published reports [27–29]. The PCOM shows the phase morphology evolution during the annealing at melt state, while SEM images (lower row of Fig. 1) displays the phase structure at room temperature after the annealing was conducted at melt state for various times. Based on Fig. 1, it was found that the SEM images showed similar phase morphology as the PCOM did, indicating that the cooling down to room temperature didn't change significantly the phase morphology evolution during the annealing. It is noted here the reason why PS60/PMMA40 was chosen for this study is that the dispersed phase domains (PMMA phase) can grow from less than 2 μm before annealing to around 15 μm after 100-min annealing, which is a much wider range of the phase domain size than that of PS20/PMMA80 and PS80/PMMA20 based on the previous work [30]. Therefore, it is expected that the relationship between the phase structure of PS60/PMMA40 blends controlled by phase separation and the foaming behavior can be studied on a wider scale.

To investigate the phase morphology evolution on the foaming behavior of the PS60/PMMA40 blend, the domain size, domain density and the interface area are shown in Fig. 2. The domain size was the average diameter of at least 100 phase domains, and the domain density was the number of the phase domains per unit volume, statistically measured with the SEM images according to the previous study [5]. The interface area was calculated with the domain size and domain density. With the increase of the annealing time, the average domain size was increased from 1.69 μm to 10.97 μm, while the domain density was sharply decreased from 1.43×10^{11} domains/cm³ to 3.54×10^8 domains/cm³, and the interface area was reduced from 12852 cm²/cm³ to 1337 cm²/cm³. Based on the previous study [31], the increase of phase domains leads to the decrease of interface areas and hence the weakened heterogeneous cell nucleation. Thereby, it is expected that the annealing in this study would significantly affect the foaming behavior.

3.2. Effect of structural evolution on foaming behavior

The cell structures of PS60/PMMA40 blend foams prepared at 130 °C after annealing for various times and a subsequent saturation at 1.5, 2, 3, 3.5 and 4 MPa for 48 h are shown in Fig. 3. It can be seen that for the same saturation pressure, different types of cell structures were observed with the prolongation of the annealing time. For the saturation pressure of 1.5 MPa, only the blend sample with 2-min thermal annealing showed uniform cell structure, with longer annealing, non-foamed particles were observed in the well foamed PS matrix. For all other saturation pressures, for samples with annealing for 2 min, 7 min or 15 min, uniform cell structures were observed, with cell size about 5–6 μm. However, for those with annealing for 30 min, 60 min or 100 min, heterogeneity in cell structure was observed. After saturation under 2 MPa, the petal-like cell structure was observed with annealing for more than 30 min. The longer the annealing proceeded, and the bigger the core part was. For the saturation at 3 MPa, the petal-like cell structure was observed with the annealing for 30 min and 60 min, while with 100-min annealing, the dispersed phase was detached from the matrix. For the saturation at 3.5 MPa, similar cell structure was observed as that for saturation at 3 MPa, while the foamability of the blend sample turned even worse with 100-min annealing. For the saturation at 4 MPa, the petal-like cell structure was observed for 30-min annealing, while the nonfoamed dispersed phase was detached from the matrix both for 60-min and 100-min annealing. The results above indicated that a longer thermal annealing time could impact significantly the foaming behavior of the PS60/PMMA40 blend. Based on the previous study [32–34], for immiscible polymer blends, the interfacial tension would remain unchanged during the thermal annealing, and therefore, the increase of the phase domain size has been proven an important factor in impacting the foaming behavior. Further, a

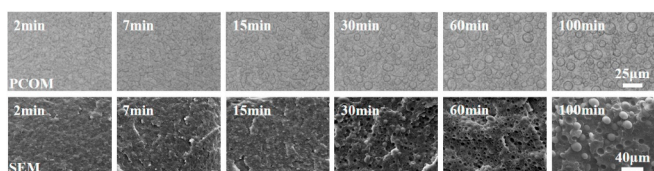


Fig. 1. PCOM and SEM images of PS60/PMMA40 blends annealing at 220 °C for various times.

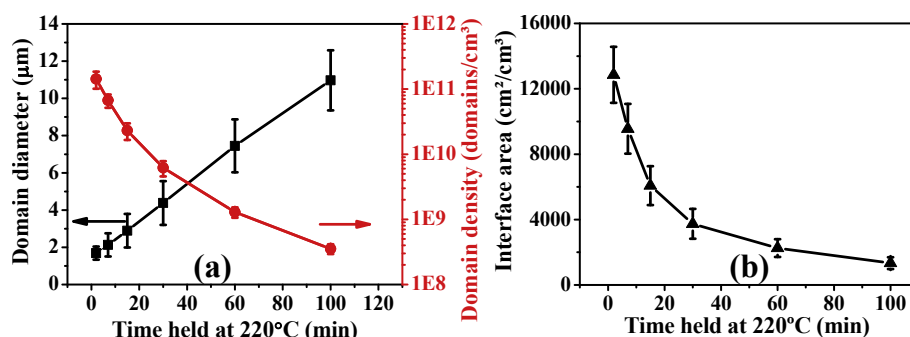


Fig. 2. (a) The domain diameter and domain density, and (b) the interface area per volume of PS60/PMMA40 blends after annealing for various times.

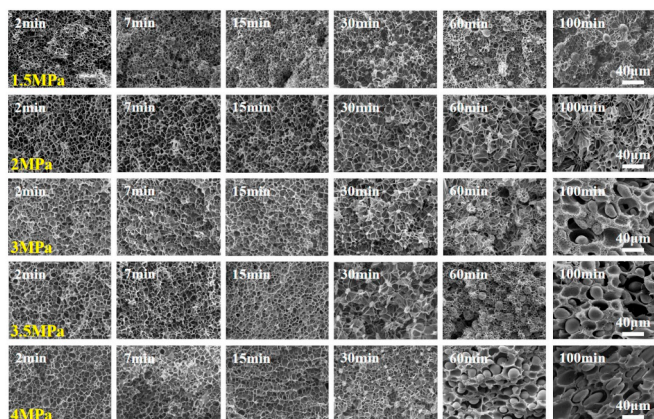


Fig. 3. Cell structures of PS60/PMMA40 blend foams prepared at 130 °C for 15 s after saturation under 1.5, 2, 3, 3.5 and 4 MPa for 48 h.

combined higher saturation pressure would make the impact even more obvious. The foaming process after the saturation under a higher pressure could detach the dispersed phase from the matrix. The higher the pressure was, and the less annealing time was required to detach the interface.

To identify the dispersed phase and the matrix in the blend foams, energy dispersive spectrum (EDS) was conducted with the EVO18 SEM. The EDS of PS60/PMMA40 blend foams prepared with an annealing for 100 min and a subsequent saturation under 2 MPa and 3 MPa is shown in Fig. 4a and Fig. 4b, respectively. In Fig. 4a, as the core part (spectrum 1) contains 75.85 wt.% C and 24.15 wt.% O, and the petal-like part (spectrum 2) nearly 100 wt.% C, indicating that the core part is the PMMA rich phase and the petal-like part the PS rich phase. Similarly, the EDS data in Fig. 4b also illustrate that the detached dispersed phase is the PMMA rich phase, and the matrix the PS rich phase. It can be seen that when the phase domain before foaming was relatively small (i.e., annealing for 2–15 min), uniform cell structure was formed, with cell

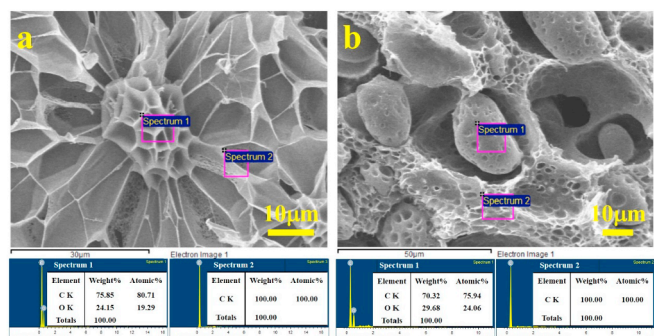


Fig. 4. Energy spectrum scheme of PS60/PMMA40 blend foams after annealing at 220 °C for 100 min and saturated under (a) 2 MPa and (b) 3 MPa for 48 h.

size larger than the original phase domain. In comparison, when the phase domain was relatively large (i.e., annealing for 30–100 min), heterogeneity in cell structure occurred, and the cell size was smaller than the phase domain, thus making the different foaming characteristics of the two components be observed. In specific, the petal-like cell structure in Fig. 4a seemed to be formed after a moderate gas transfer from PMMA phase to PS phase, leaving gas transfer tracks surrounding the gas reservoir, while the cell structure in Fig. 4b seemed to be formed after a more forceful gas transfer, thus detaching the interface. These results indicate that for the blend samples with big phase domains, as the saturation pressure increases, the dispersed PMMA phase gradually cannot be able to coordinate well with the PS matrix during foaming, probably due to the viscoelasticity difference.

The documented studies [35,36] reported that the small cell domains with large interfacial areas are favorable for formation of uniform cell structures ascribed to the strong heterogeneous cell nucleation. Our previous work [30] found that with the increase in phase domain size, the heterogeneous cell nucleation is weakened, while the homogeneous cell nucleation is gradually not insignificant any more. Sharudin et al. [36] reported that the nonfoamed PP dispersed phase is detached from the PS matrix during foaming due to the stiffness of PP when PP is hardly foamed at a lower foaming temperature. Gong et al. [31] also reported that the PET particles are detached from the PC matrix during foaming when the blends without interfacial compatibilization were foamed at a lower foaming temperature that is hard for PET to foam. Taki et al. [37] observed that the large and open cells embracing nonfoamed PEG particles were surrounded by small PS cells in PEG/PS blends due to bubble coalescence in PEG phase ascribed to its low modulus. These findings [31,35–37] point to the deduction that the difference in viscoelasticity in binary polymer blends has proven an important factor in determining the detachment of the dispersed phase from the matrix during foaming.

3.3. Effect of structural evolution on desorption diffusion coefficient

As the petal-like cell structure or even detachment at interface was observed for PS60/PMMA40 blend sample after a relatively long annealing time, it is necessary to study the effect of the structural evolution on the gas diffusion during desorption. In addition, according to the previous studies [6,7,38], the desorption diffusion coefficient is an important factor in evaluating the gas escape during foaming and thus affecting the expansion ratio. The desorption curves and the desorption diffusion coefficients of pure PS, pure PMMA and the PS60/PMMA40 blends after annealing for various times and saturation under 3 MPa for 48 h are shown in Fig. 5. It was observed in Fig. 5a that the diffusion curves were linear lines for pure PS and pure PMMA, while they were not linear for the blend samples. This indicated that the gas diffusion didn't follow the Fickian law for the latter ones. The diffusion curves of the blend samples can be divided into three parts: the beginning part (Fig. 5b), the middle part and the end part (Fig. 5c) according to the previous literature [6]. The beginning part and the end part are related

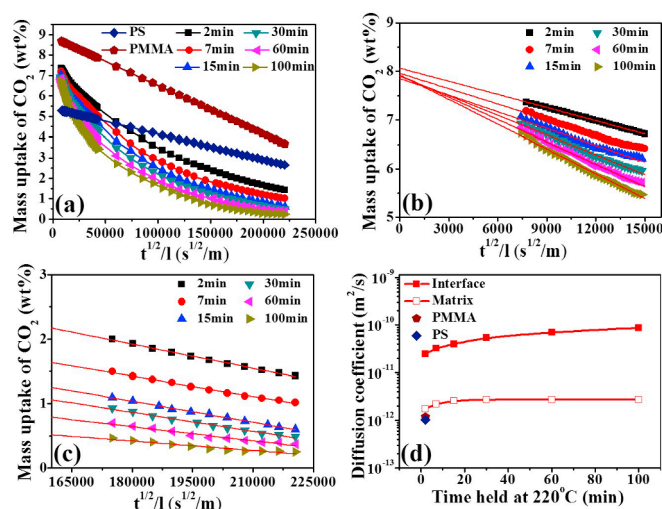


Fig. 5. (a) The desorption kinetics of pure PS, pure PMMA and PS60/PMMA40 blends after saturation under 3 MPa for 48 h, with (b) the magnified beginning part, (c) the magnified end part, and (d) the corresponding desorption diffusion coefficients.

to the gas diffusion from the interface and matrix, respectively. The desorption diffusion coefficient of pure PS, pure PMMA and the interface and matrix of PS60/PMMA40 blend can be obtained based on equation (2) [21], and is shown in Fig. 5d. The desorption diffusion coefficient was about $1.04 \times 10^{-12} \text{ m}^2/\text{s}$ and $1.19 \times 10^{-12} \text{ m}^2/\text{s}$ for pure PS and PMMA, respectively. It was increased from $2.47 \times 10^{-11} \text{ m}^2/\text{s}$ to $8.69 \times 10^{-11} \text{ m}^2/\text{s}$ for the interface of PS60/PMMA40 blends with the annealing from 2 min to 100 min, and from $1.74 \times 10^{-12} \text{ m}^2/\text{s}$ to $2.78 \times 10^{-12} \text{ m}^2/\text{s}$ for the matrix. As the interface surrounding each PMMA phase domain was increased with the annealing, while the total interface was decreased, the gas diffusion channel was shortened, which allowed easier gas escape. Hence, the desorption diffusion coefficient for the interface was increased with the annealing. Zhai et al. [6] reported that the gas diffusion coefficient of PP/PS blends with co-continuous phase structure was decreased with interfacial adhesion upon compatibilization.

3.4. Effect of structural evolution and saturation pressure on expansion ratio

As the desorption diffusion coefficient for the interface was increased with the annealing due to the increase of phase domains, it thus can be imagined that the expansion ratio would be significantly affected based on the previous studies [6,7,38]. The expansion ratio of the PS60/PMMA40 blend sample was studied as a function of the annealing time for various saturation pressures, as shown in Fig. 6. Several observations are worth being mentioned as follows. First, for the samples annealing for 2 min, the expansion ratio was the highest with a saturation pressure of 3 MPa. In comparison, the lower solubility of CO_2 would be too low for the sample to be well foamed after saturation under 1.5 MPa, and it was still relatively low after saturation under 2 MPa compared to that under 3 MPa. Whereas, after saturation under a pressure higher than 3 MPa, the expansion ratio was decreased again, probably due to the decreased viscosity upon the strong plasticization effect of CO_2 [39]. Thus, 3 MPa was probably a suitable saturation pressure for the sample annealing for 2 min to be well foamed at 130°C . Second, for the samples annealing for 100 min, the expansion ratio was decreased with the saturation pressure. The higher the saturation pressure was, and the lower the expansion ratio was. This was believed to be related to a larger extent of detachment during foaming after saturation under a higher pressure, although the samples didn't show obvious detachment after saturation under 1.5 MPa and 2 MPa. Third,

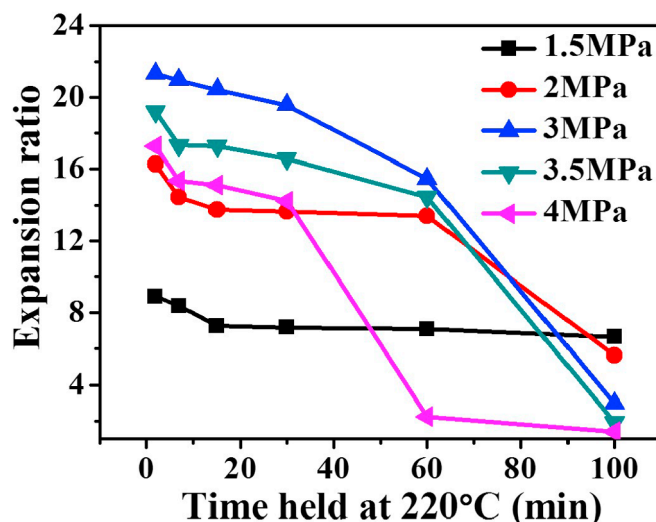


Fig. 6. The expansion ratio of PS60/PMMA40 blend foams prepared after annealing for various times and saturation under different pressures.

the expansion ratio was decreased with the annealing time for all the saturation pressures, which was due to the easier gas escape related to the increase of the desorption diffusion coefficient. However, the decrease in expansion ratio was varied with the saturation pressure. For each saturation pressure at 1.5, 2, 3, 3.5 and 4 MPa, the decrease for annealing from 2 min to 100 min was 2.25, 10.65, 18.36, 17.29, and 15.85, respectively. This illustrates that the reduction in expansion ratio was the largest for the sample saturated under 3 MPa. This indicates that the foaming condition at 130°C for 15 s was most suitable for the blend sample with an annealing for 2 min followed by the saturation under 3 MPa. This further indicates that a suitable foaming temperature range would be required for each sample with a certain annealing time and saturated under a certain pressure.

3.5. Dependence of foaming window on phase domain and saturation pressure

Whether the phase domain size and the saturation pressure would be important factors in affecting the foaming temperature window of the PS60/PMMA40 blend sample still remains an unsolved question. To address this issue, various foaming temperatures were applied to study on the foaming behavior for the blend samples annealed for different times followed by a subsequent saturation process under certain pressure. The upper (T_U) and lower (T_L) limits of the foaming temperatures would thus be determined for each saturation pressure, and hence the suitable foaming temperature window. Here, to be more readable, only the suitable foaming temperature range for the samples annealed for 2 min and 100 min will be presented. Based on the previous studies [40,41], the foaming temperature window is often investigated for linear and long-chain branched PP in continuous extrusion foaming. However, the upper and lower temperature limits are usually not exactly defined by definite boundaries with cell structure parameters. Resultantly, the suitable foaming temperature window is usually obtained as a rough range for the samples to be well foamed with no less than a certain expansion ratio and with no thick cell walls. In this study, as it is expected to study the dependence of the foaming temperature window on the phase domain size and the saturation pressure, hence, the trend of foaming window as a function of the two factors is the key point, instead of the exact temperature limits. Thereby, the expansion ratio of more than 3 with no thick cell walls are used in this study to define the well foamed samples, and thus to obtain the suitable foaming temperature range.

The cell structures of PS60/PMMA40 blend foams with annealing for 2 min and 100 min and a subsequent saturation under 1.5 MPa are

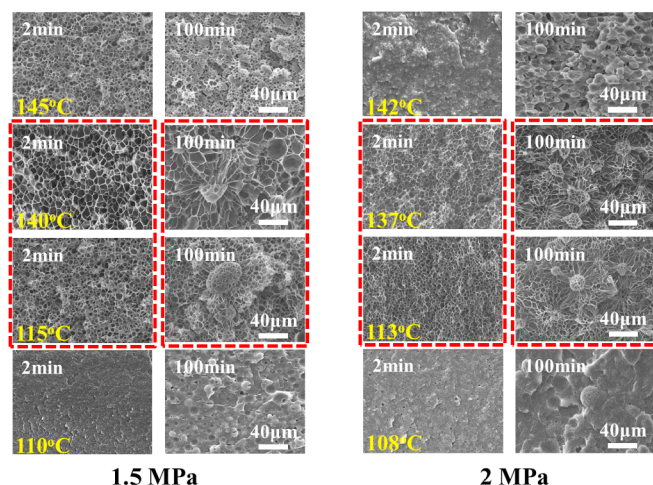


Fig. 7. Determination of the suitable foaming temperature range for PS60/PMMA40 blend samples after saturation under 1.5 MPa (left) and 2 MPa (right). The dashed squares represent the suitable foaming temperature range for the samples annealing for 2 min and 100 min, respectively.

shown in Fig. 7 (the left two columns). To determine the T_U and T_L for each saturation pressure, the foaming temperature interval of 5 °C was applied to distinguish the cell structures. Based on the previous studies [37,40–44], the upper and lower limit of the foaming temperature is generally determined by the melt strength and the stiffness of the polymer system, respectively. For the PS60/PMMA40 blend with annealing for 2 min, the sample can be well foamed at 140 °C, but not at 145 °C, with a very low expansion ratio only about 1–3 and very thick cell walls at the latter temperature due to the cell shrinkage and collapse. If the sample had been foamed at a higher temperature like 146 °C, very few cells would have still existed in the samples. Therefore, the T_U should be between 145 °C and 140 °C. In addition, the sample can be foamed at 115 °C, but not at 110 °C due to the stiffness of the samples. Hence, the T_L was determined between 115 °C and 110 °C. For the PS60/PMMA40 blend with annealing for 100 min, both the T_U and the T_L were similar as those of the blend annealed for 2 min. This indicates that after a saturation process under lower pressures, the suitable foaming temperature range is similar for the blend with a short and a long annealing process. However, compared to the blend with small dispersed phase domains, the heterogeneity was observed in cell structure in the blend with big phase domains. Non-foamed PMMA phase domains were observed at the foaming temperature of 145 °C, and different foaming characteristics were found between the dispersed phase and the matrix at the foaming temperatures of 140 °C and 115 °C. Combined this finding with the cell structure observation in Fig. 3, it is expected that after saturation under higher pressures, the difference between the cell structures of the blend with a short and a long annealing time could be more prominent.

The cell structures of PS60/PMMA40 blend foams after the saturation under the pressure of 2 MPa are shown in Fig. 7 (the right two columns). It can be seen that for the blend sample annealed for 2 min, the T_U was between 142 °C and 137 °C, and the T_L was between 113 °C and 108 °C. For the sample with a prior annealing process for 100 min, both the T_U and T_L were similar as those of the blend annealed for 2 min. This indicates that the big phase domain induced by phase separation might not be able to alter the suitable foaming temperature range of the blend after saturation under a relatively low pressure, although heterogeneity was observed in cell structure. Based on the reported documents [45,46], the plasticization effect upon dissolving CO₂ in polymers would bring in reduction in both melt strength and stiffness, and this phenomenon is more prominent with higher amounts of CO₂ dissolved. Therefore, it is expected that after the blend sample was saturated under higher pressures, both the T_U and the T_L would be

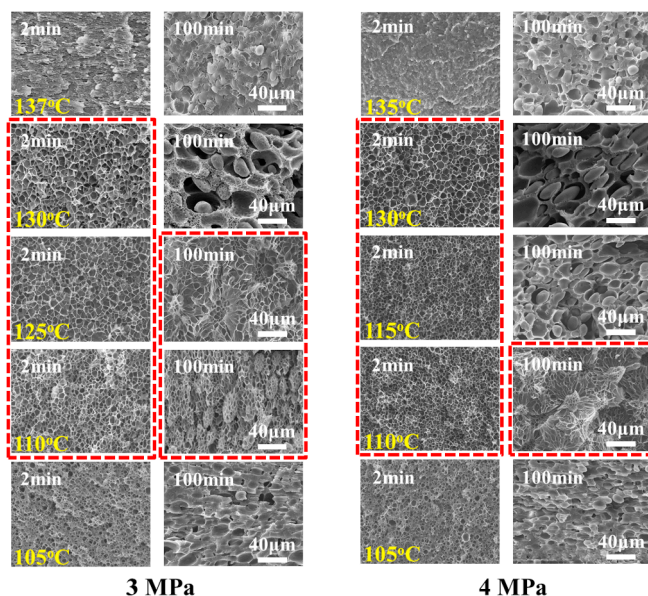


Fig. 8. Determination of the suitable foaming temperature range for PS60/PMMA40 blend samples after saturation under 3 MPa (left) and 4 MPa (right). The dashed squares represent the suitable foaming temperature range for the samples annealing for 2 min and 100 min, respectively.

depressed, although the decrease could be in various degrees, depending on the saturation pressure, the viscoelasticity of polymers and other factors based on the previous reports [42–46]. Compared to the T_U and T_L for the blend sample saturated under 1.5 MPa, those of the counterparts under 2 MPa were decreased, which confirmed the conjecture above.

The cell structures of PS60/PMMA40 blend foams prepared under a saturation pressure of 3 MPa are presented in Fig. 8 (the left two columns). For the blend sample annealed for 2 min, the T_U was between 137 °C and 130 °C, and the T_L was between 110 °C to 105 °C. In comparison, for the sample with annealing for 100 min, they were between 130 °C and 125 °C, and between 110 °C to 105 °C, respectively. The T_L was still the same for the two samples, while the T_U was decreased for the latter. Unlike the samples saturated under 1.5 MPa and 2 MPa, the suitable foaming temperature range of the one annealed for 100 min and saturated under 3 MPa was no longer similar as that of the counterpart annealed for 2 min, but instead, it turned narrower. This meant that the T_U was more sensitive to the saturation pressure increase than the T_L was, and this phenomenon was more prominent for the blend sample with large phase domains. It further indicates that a larger phase domain size in this study is more vulnerable to a combination of a higher saturation pressure and a higher foaming temperature. Take the blend sample annealed for 100 min, saturated under 3 MPa and foamed at 130 °C for an example, the dispersed phase and the matrix was detached during foaming, resulting in very bad foamability. To make the sample be well foamed without detachment at the interface, either a lower saturation pressure or a lower foaming temperature was required.

The cell structures of the PS60/PMMA40 blend foams undergoing a saturation process at 4 MPa are shown in Fig. 8 (the right two columns). For the blend sample annealed for 2 min, the T_U was determined between 135 °C and 130 °C, and the T_L between 110 °C and 105 °C. However, for the blend sample annealing for 100 min, although the T_L was also between 110 °C and 105 °C, same as that annealing for 2 min, the T_U was between 115 °C and 110 °C, much lower than that annealing for 2 min, making the suitable foaming temperature range quite narrow. Compared to that of the blend sample annealing for 100 min and saturated under 3 MPa, the suitable temperature range turned even narrower for the sample saturated under 4 MPa. This indicates that the T_U for the blend with very large phase domains is more significantly

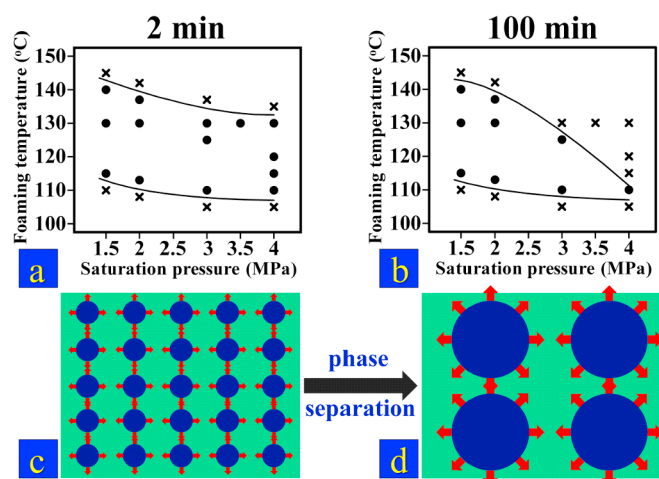


Fig. 9. The dependence of the suitable foaming temperature window on the saturation pressure for PS60/PMMA40 blend samples after annealing for (a) 2 min and (b) 100 min, respectively. The corresponding schematics for interpreting the different dependences for the samples with (c) small and (d) big phase domains. (● can be well foamed; × cannot be well foamed).

affected by the saturation under higher pressures. This can further be confirmed by the comparison of the foaming behavior and the relevant cell structure after saturation under 3 MPa and 4 MPa as follows. After the blend sample was annealed for 100 min, saturated under 3 MPa and foamed at 130 °C, the interface was detached during the foaming, while the alteration of the preparation conditions above into a shorter annealing time of 60 min (Fig. 3) or a lower foaming temperature of 125 °C prevented the interface detachment (Fig. 8). In comparison, the interface was detached not only for the blend sample annealed for 100 min, saturated under 4 MPa and foamed at 130 °C, but also for those counterparts annealed for 60 min (Fig. 3) or foamed at 115 °C (Fig. 8).

Based on the results above, the suitable foaming temperature range as a function of the saturation pressure of the PS60/PMMA40 annealed for 2 min (a) and 100 min (b) and the corresponding schematics (c and d) are shown in Fig. 9. Based on Fig. 9a and b, several observations were worthy being mentioned. First, both the T_U and the T_L were decreased with the saturation pressure from 1.5 MPa to 4 MPa, while the former was more sensitive to the pressure increase. This trend applied to both samples. Second, the T_L as a function of the saturation pressure almost overlapped for the blend samples with a short and a long annealing time, while the T_U was just similar for the two samples saturated under lower pressures, and was much more vulnerable to the pressure increase for the 100-min annealing sample. Third, the suitable foaming temperature window for the blend sample annealing for 2 min turned a little narrower from 30 °C to 25 °C with the pressure increase, while the temperature window for the sample annealing for 100 min gradually became much narrower from 30 °C to 5 °C. These observations have proven that the phase domain size and the saturation pressure are important factors in altering the upper foaming temperature for the blend sample in this study.

To interpret the observations above, the schematics about the gas transfer and escape during foaming of binary immiscible polymer blends with small and big phase domains are presented in Fig. 9c and d, respectively. First, based on the previous study [46], the glass transition temperature (T_g) is decreased due to the plasticization effect of polymers upon dissolving CO_2 . Thereby, both the melt strength and the stiffness of the polymers would be depressed with the saturation pressure, which led to the gradual decrease in both T_U and T_L , regardless of the dimension of the phase domain. Second, both the two samples were too stiff to foam at very low foaming temperatures, no matter what the phase domain size was, and therefore, the T_L almost overlapped for the

two blend samples under various pressures. In comparison, the determination of T_U was more complicated. For the sample annealing for 2 min, the melt strength drop due to the plasticization effect of CO_2 would be the dominant factor in determining the trend of T_U as a function of the saturation pressure. However, for the sample annealing for 100 min, although the T_U after saturation under lower pressures was determined by the melt strength of polymers, it was significantly affected by the domain size after saturation under higher pressures. This was because the difference in viscoelasticity between the dispersed phase and the matrix was increased to a large extent, and thus exerted a more prominent effect on the foaming behavior, which was interpreted as follows. PMMA phase has a larger solubility of CO_2 than PS phase does due to the higher affinity of CO_2 with the carbonyl group [47,48]. Based on the previous report [46], the more CO_2 dissolves into the polymers, and the more prominently T_g is decreased. According to other document [49], the T_g of PMMA is a little higher than that of PS without dissolving CO_2 , while after dissolving a large amount of CO_2 under a higher saturation pressure, it is decreased more obviously for PMMA than for PS. Therefore, after saturation under higher pressures, the viscosity of PMMA phase would be too low to hold the gas. During foaming, the gas in PMMA phase would be inclined to diffusing to the interface and PS phase. The previous literature also reported that PMMA phase can work as a CO_2 reservoir [9]. In this study, for a small phase domain, the gas transfer of a small amount of the gas couldn't be able to detach the interface. However, for a big phase domain, the interface could be detached. This could possibly be explained by the Laplace pressure [50], which varies proportionally to the interfacial tension and inversely with the domain size. For the blend samples with bigger phase domains, the Laplace pressure would be smaller, and thus couldn't prevent the gas transfer from the dispersed phase to the interface and the matrix. Once the interface was detached, the CO_2 gas was escaped from the interface very quickly, resulting in bad foamability of blend sample. To get the blend sample be well foamed after a saturation process under higher pressures, the upper foaming temperature needed to be lowered. Hence, the foaming window was very narrow for the PS60/PMMA40 blend sample annealed for 100 min.

4. Conclusions

The dependence of the foamability and the foaming window of PS60/PMMA40 blend was thoroughly studied on the structural evolution driven by phase separation with thermal annealing as a feasible approach to precisely tune the phase domain. To the best of our knowledge, this is the first time to report the dependence of the foaming window on the phase domain size. It is found that the phase domain is increased with the thermal annealing, and the interface is gradually decreased. As a result, the cell structure is uniform for the blend samples with short-period annealing, while heterogeneity is observed for those with long-period annealing, and even detachment at the interface for those with long-period annealing combined with high-pressure saturation. This indicates that the phase domain size in this study plays an important role in affecting the foaming behavior of the blend samples. The gas diffusion coefficient is found to increase with the domain size, ascribed to the shortened distance for gas escape, resulting in the decreased expansion ratio. Although the expansion ratio shows the same decreasing trend versus the annealing time for each saturation pressure, the reduction is varied with the saturation pressure. It indicates that the suitable foaming temperature would be required for each of the blend samples to be well foamed after annealing and saturation under different conditions. Hence, the suitable foaming temperature range was studied for the blend samples with different annealing times followed by saturation under each pressure, and the foaming temperature window was obtained for the blend samples with a short and a long annealing period, respectively. It is found that both the upper and the lower foaming temperature are decreased with the saturation pressure, and the upper one is more sensitive to the higher saturation pressures

for the sample with a long period of annealing, making the relevant suitable foaming temperature window much narrower. It is believed that the difference in viscoelasticity between the two components is responsible for that, and would be more prominent in affecting the foaming behavior of the blend samples with large phase domains and saturated under higher pressures.

Based on this study, it is believed that the preparation condition of a big phase domain, a high saturation pressure and a high foaming temperature could be unfavorable for the PS60/PMMA40 blend to be well foamed. With two of them kept unchanged, an increase in the third one could lead to the heterogeneity in cell structures, or even detachment at the interface. To get the blend sample be well foamed, a smaller phase domain, a slightly lower saturation pressure, or a slightly lower foaming temperature could be required.

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